



1. Description

1.1. Project

Project Name	Tracking
Board Name	custom
Generated with:	STM32CubeMX 6.15.0
Date	03/04/2026

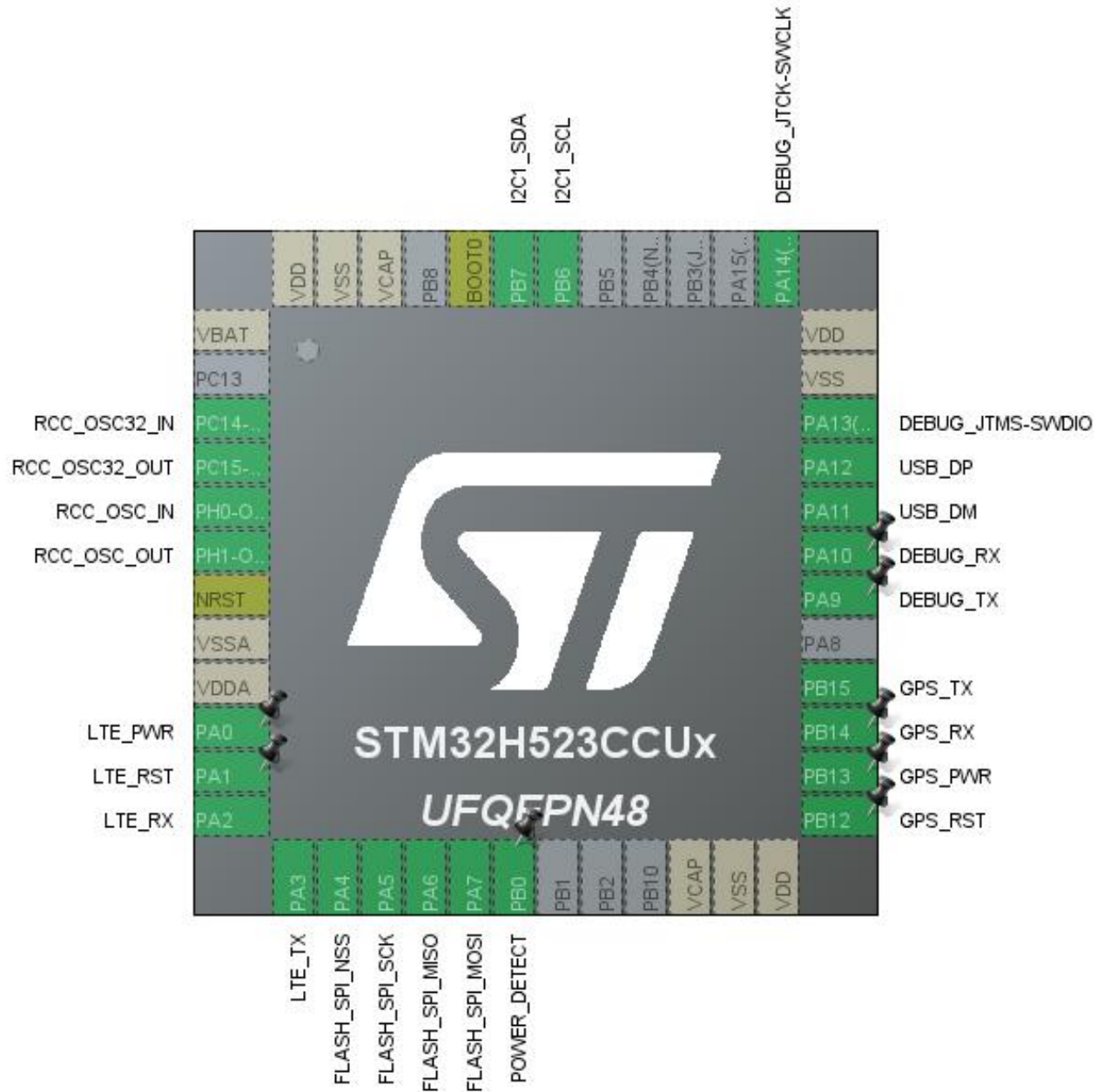
1.2. MCU

MCU Series	STM32H5
MCU Line	STM32H5x3
MCU name	STM32H523CCUx
MCU Package	UFQFPN48
MCU Pin number	48

1.3. Core(s) information

Core(s)	ARM Cortex-M33
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2. Pinout Configuration



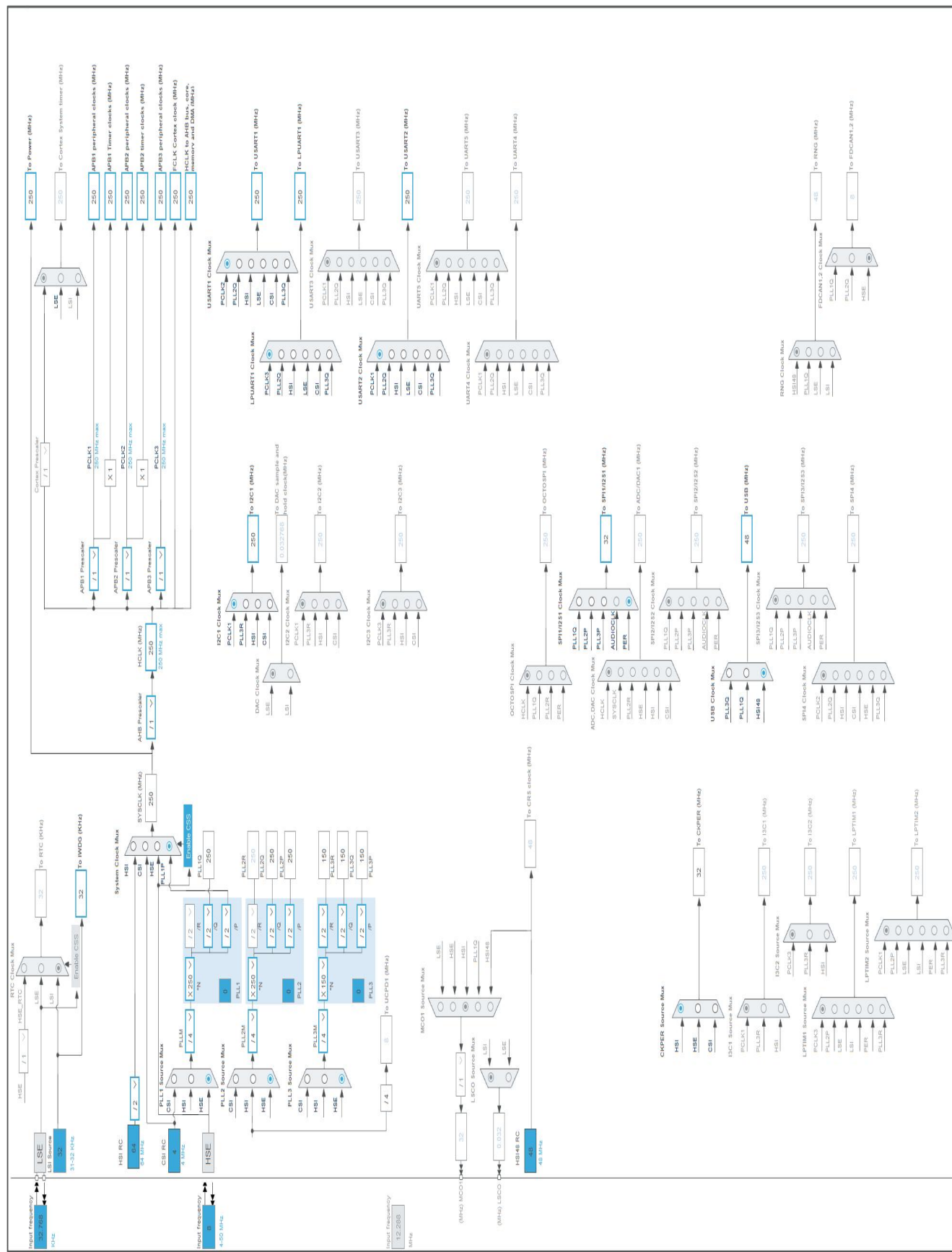
3. Pins Configuration

Pin Number UFQFPN48	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
3	PC14- OSC32_IN(OSC32_IN)	I/O	RCC_OSC32_IN	
4	PC15- OSC32_OUT(OSC32_OUT)	I/O	RCC_OSC32_OUT	
5	PH0-OSC_IN(PH0)	I/O	RCC_OSC_IN	
6	PH1-OSC_OUT(PH1)	I/O	RCC_OSC_OUT	
7	NRST	Reset		
8	VSSA	Power		
9	VDDA	Power		
10	PA0 *	I/O	GPIO_Output	LTE_PWR
11	PA1 *	I/O	GPIO_Output	LTE_RST
12	PA2	I/O	USART2_TX	LTE_RX
13	PA3	I/O	USART2_RX	LTE_TX
14	PA4	I/O	SPI1_NSS	FLASH_SPI_NSS
15	PA5	I/O	SPI1_SCK	FLASH_SPI_SCK
16	PA6	I/O	SPI1_MISO	FLASH_SPI_MISO
17	PA7	I/O	SPI1_MOSI	FLASH_SPI_MOSI
18	PB0	I/O	GPIO_EXTI0	POWER_DETECT
22	VCAP	Power		
23	VSS	Power		
24	VDD	Power		
25	PB12 *	I/O	GPIO_Output	GPS_RST
26	PB13 *	I/O	GPIO_Output	GPS_PWR
27	PB14	I/O	USART1_TX	GPS_RX
28	PB15	I/O	USART1_RX	GPS_TX
30	PA9	I/O	LPUART1_TX	DEBUG_TX
31	PA10	I/O	LPUART1_RX	DEBUG_RX
32	PA11	I/O	USB_DM	
33	PA12	I/O	USB_DP	
34	PA13(JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	
35	VSS	Power		
36	VDD	Power		
37	PA14(JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	
42	PB6	I/O	I2C1_SCL	
43	PB7	I/O	I2C1_SDA	

Pin Number UFQFPN48	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
44	BOOT0	Boot		
46	VCAP	Power		
47	VSS	Power		
48	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H5
Line	STM32H5x3
MCU	STM32H523CCUx
Datasheet	DS00000 Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

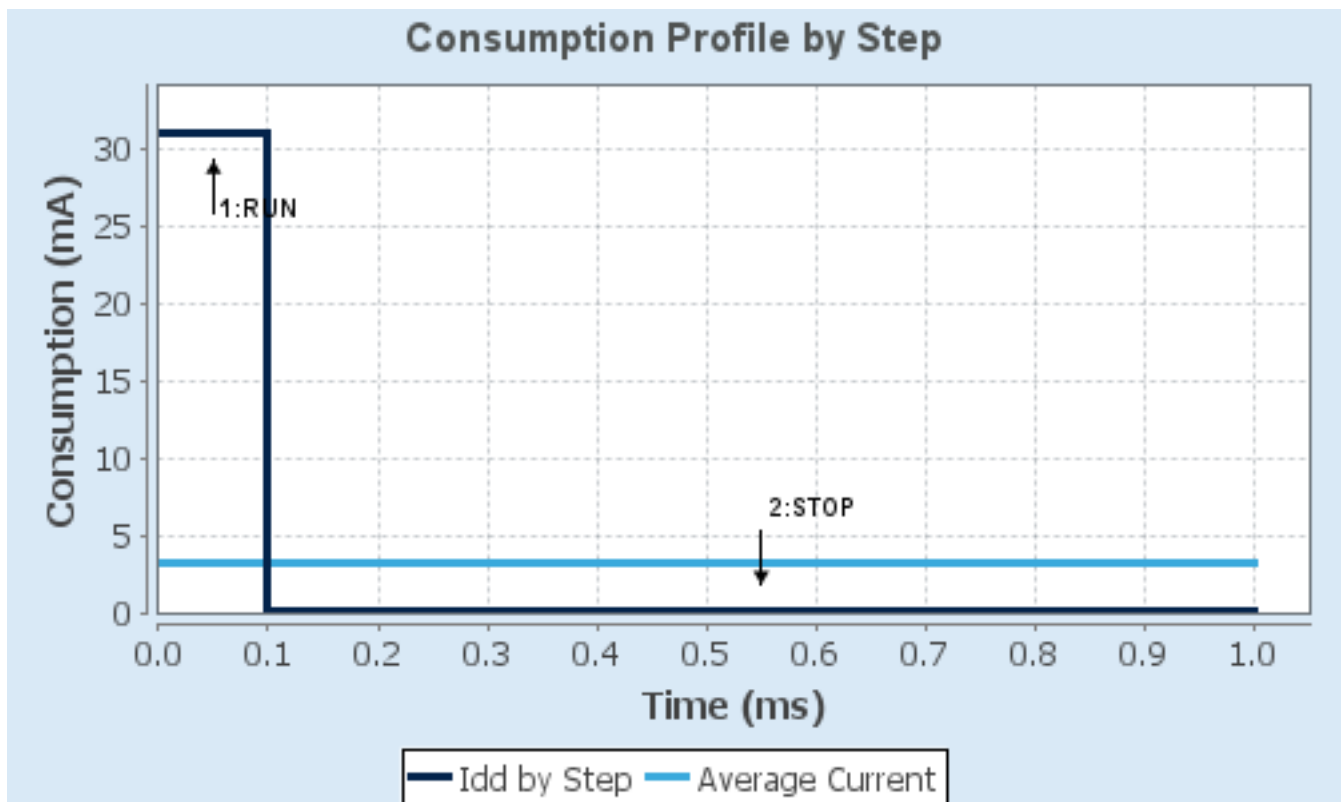
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0	SVOS5: System-Scale5
Fetch Type	FLASH_ON/Cache2Ways_A LL_RAM_RETENTION	FLASH_PwrDwnStop_OFF
CPU Frequency	250 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL_CLOCKS_OFF
Clock Source Frequency	8 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	31 mA	100 μ A
Duration	0.1 ms	0.9 ms
DMIPS	375.0	0.0
Ta Max	125	125
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	3.19 mA
Battery Life	1 month, 13 days, 22 hours	Average DMIPS	37.5 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	Tracking
Project Folder	D:\SynaptiX\Firmware\Tracking
Toolchain / IDE	Makefile
Firmware Package Name and Version	STM32Cube FW_H5 V1.5.1
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DCACHE1_Init	DCACHE1
4	MX_I2C1_Init	I2C1
5	MX_ICACHE_Init	ICACHE
6	MX_IWDG_Init	IWDG
7	MX_LPUART1_UART_Init	LPUART1
8	MX_USART1_UART_Init	USART1
9	MX_USART2_UART_Init	USART2
10	MX_SPI1_Init	SPI1
11	MX_USB_PCD_Init	USB

3. *Peripherals and Middlewares Configuration*

3.1. BOOTPATH

mode: Activated

3.2. DCACHE1

mode: Activated

3.2.1. Parameter Settings:

Basic Parameters:

DCACHE Read Burst Type

DCACHE READ BURST WRAP

3.3. DEBUG

Debug: Serial Wire

3.4. I2C1

I2C: I2C

3.4.1. Parameter Settings:

Timing configuration:

I2C Speed Mode

Fast Mode Plus *

I2C Speed Frequency (KHz)

1000

Rise Time (ns)

0

Fall Time (ns)

0

Coefficient of Digital Filter

0

Analog Filter

Enabled

Timing

0x00C035A6 *

Slave Features:

Clock No Stretch Mode

Disabled

General Call Address Detection

Disabled

Primary Address Length selection

7-bit

Dual Address Acknowledged

Disabled

Primary slave address

0

3.5. ICACHE

mode: Memory address remap

3.5.1. Parameter Settings:

Region 0:

Region	Disable
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Region 1:

Region	Disable
--------	---------

Region 2:

Region	Disable
--------	---------

Region 3:

Region	Disable
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3.6. IWDG

mode: Activated

3.6.1. Parameter Settings:

Watchdog Clocking:

IWDG counter clock prescaler	64 *
IWDG window value	4095
IWDG down-counter reload value	4095

Watchdog Interrupt:

IWDG Early Wakeup Interrupt	0
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3.7. LPUART1

Mode: Asynchronous

3.7.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Single Sample	Disable
ClockPrescaler	1

Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.8. MEMORYMAP

mode: Activated

3.9. PWR

mode: Power saving mode

mode: Privilege attributes

3.9.1. PWR Privilege :

Privilege PWR:

PWR Privilege	Disable
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3.10. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

Low Speed Clock (LSE) : Crystal/Ceramic Resonator

3.10.1. RCC Privilege :

Privilege RCC:

Privilege of RCC Non-Secure Items	Disable
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3.10.2. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	5 WS (6 CPU cycle)

Flash Programming Delay 2

RCC Parameters:

HSI Calibration Value 64

CSI Calibration Value 32

HSE Startup Timeout Value (ms) 100

LSE Startup Timeout Value (ms) 5000

TIM Prescaler Selection Disabled

Power Parameters:

Power Regulator Voltage Scale Power Regulator Voltage Scale 0

PLL1/2/3 Parameters:

PLL1 input frequency range Between 2 and 4 MHz

3.11. SPI1

Mode: Full-Duplex Master

Hardware NSS Signal: Hardware NSS Output Signal

3.11.1. Parameter Settings:

Basic Parameters:

Frame Format Motorola

Data Size **8 Bits ***

First Bit MSB First

Clock Parameters:

Prescaler (for Baud Rate) **4 ***

Baud Rate **8.0 MBits/s ***

Clock Polarity (CPOL) Low

Clock Phase (CPHA) 1 Edge

CRC Parameters:

CRC Calculation Disabled

Advanced Parameters:

NSSP Mode Enabled

NSS Signal Type Output Hardware

Fifo Threshold Fifo Threshold 01 Data

Nss Polarity Nss Polarity Low

Master Ss Idleness 00 Cycle

Master Inter Data Idleness 00 Cycle

Master Receiver Auto Susp Disable

Master Keep Io State Master Keep Io State Disable

IO Swap Disabled

Ready Master Management Internal

Ready Signal Polarity

High

3.12. SYS

Timebase Source: SysTick

3.13. USART1

Mode: Asynchronous

3.13.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.14. USART2

Mode: Asynchronous

3.14.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.15. USB

Mode: Device__Only

3.15.1. Parameter Settings:

Basic Parameters:

Speed	Full Speed 12MBit/s
Physical interface	Internal Phy
Signal start of frame	Disabled

Power Parameters:

Low Power	Disabled
Link Power Management	Disabled
Battery Charging	Disabled

EndPoint Parameters:

Bulk double buffer	Disabled
Iso single buffer	Disabled

3.16. AL94.I-CUBE-USB-D-COMPOSITE.1.0.3

mode: USBJComposite

3.16.1. Parameter Settings:

Basic Parameters:

USBD_USE_HS	false
USBD_USE_CDC_ACM	true *
USBD_CDC_ACM_COUNT	3 *
USBD_USE_CDC_RNDIS	false
USBD_USE_CDC_ECM	false
USBD_USE_HID_MOUSE	false
USBD_USE_HID_KEYBOARD	false
USBD_USE_HID_CUSTOM	false
USBD_USE_UAC_MIC	false
USBD_USE_UAC_SPKR	false
USBD_USE_UVC	false
USBD_USE_MSC	true *
USBD_USE_DFU	false
USBD_USE_PRNTR	false
STM32F1_DEVICE	true *

* User modified value

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
DEBUG	PA13(JTMS/SWDIO)	DEBUG_JTMS-SWDIO	n/a	n/a	n/a	
	PA14(JTCK/SWCLK)	DEBUG_JTCK-SWCLK	n/a	n/a	n/a	
I2C1	PB6	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PB7	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	
LPUART1	PA9	LPUART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	DEBUG_TX
	PA10	LPUART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	DEBUG_RX
RCC	PC14-OSC32_IN(OSC32_IN)	RCC_OSC32_IN	n/a	n/a	n/a	
	PC15-OSC32_OUT(OSC32_OUT)	RCC_OSC32_OUT	n/a	n/a	n/a	
	PH0-OSC_IN(PH0)	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT(PH1)	RCC_OSC_OUT	n/a	n/a	n/a	
SPI1	PA4	SPI1_NSS	Alternate Function Push Pull	No pull-up and no pull-down	Low	FLASH_SPI_NSS
	PA5	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	FLASH_SPI_SCK
	PA6	SPI1_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	FLASH_SPI_MISO
	PA7	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	FLASH_SPI_MOSI
USART1	PB14	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	GPS_RX
	PB15	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	GPS_TX
USART2	PA2	USART2_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	LTE_RX
	PA3	USART2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	LTE_TX
USB	PA11	USB_DM	n/a	n/a	n/a	
	PA12	USB_DP	n/a	n/a	n/a	
GPIO	PA0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LTE_PWR
	PA1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LTE_RST
	PB0	GPIO_EXTI0	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	POWER_DETECT
	PB12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPS_RST
	PB13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPS_PWR

4.2. GPDMA1

4.3. GPDMA2

4.4. LINKEDLIST

4.5. NVIC configuration

4.5.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
EXTI Line0 interrupt	true	0	0
I2C1 Event interrupt	true	0	0
I2C1 Error interrupt	true	0	0
SPI1 global interrupt	true	0	0
USART1 global interrupt	true	0	0
USART2 global interrupt	true	0	0
LPUART1 global interrupt	true	0	0
USB FS global interrupt	true	0	0
Flash non-secure global interrupt	unused		
RCC non-secure global interrupt	unused		
IWDG global interrupt	unused		
FPU global interrupt	unused		
Instruction cache global interrupt	unused		
Data cache global interrupt	unused		

4.5.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true
EXTI Line0 interrupt	false	true	true

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
I2C1 Event interrupt	false	true	true
I2C1 Error interrupt	false	true	true
SPI1 global interrupt	false	true	true
USART1 global interrupt	false	true	true
USART2 global interrupt	false	true	true
LPUART1 global interrupt	false	true	true
USB FS global interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

Middleware

Software Packs

I-CUBE-USB-D-COMP... ✓

System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Utilities	Other
CORTEX_M33			I2C1 ✓				DEBUG ✓	PWR ✓	LINKEDLIST	
DCACHE1 ✓			LPUART1 ✓							
GPDMA1			SPH ✓							
GPDMA2			USART1 ✓							
GPIO ✓			USART2 ✓							
ICACHE ✓			USB ✓							
IWDG ✓										
NVIC ✓										
RCC ✓										
SYS ✓										

6. Software Pack Report

6.1. Software Pack selected

Vendor	Name	Version	Component
AL94	I-CUBE-USBD-COMPOSITE	1.0.3	Class : USB Group : COMPOSITE SubGroup : Core Version : 1.0.0 Class : USB Group : COMPOSITE SubGroup : CDC_ACM Version : 1.0.0 Class : USB Group : COMPOSITE SubGroup : MSC_BOT Version : 1.0.0 Class : USB Group : COMPOSITE SubGroup : COMPOSITE Version : 1.0.0

7. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h5-bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h5-ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h5-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h5-series-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-entry-level-graphics.pdf
Brochures	https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf
Brochures	https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h5.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1474-security-bulletin-tn1474stpsirt-information-on-softwarebased--microarchitectural-timing-sidechannel-attacks-on-mcus-with-trustzone-for--armv8m-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf

Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-uart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5925-stm32cube-mcu-package-examples-for-stm32h5-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5927-i3c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5930-guidelines-for-power-management-on-stm32h5-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5711-getting-started-with-stm32h5-mcu-hardware-development-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5342--how-to-use-error-correction-code-ecc-management-for-internal-memories-protection-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5872-introduction-to-the-system-architecture-and-performance-in-the-stm32h5-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4894-how-to-use-eeeprom-emulation-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signaltonoise-ratio-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4992-introduction-to-

secure-firmware-install-sfi-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5405-how-to-use-fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an6014-introduction-to-sfsp-versions-for-stm32h5-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

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